$\cos^2\dfrac{A+B}{2} - \cos\dfrac{A-B}{2}\cos\dfrac{A+B}{2} + 2u = 0$, which is a quadratic in $\cos\dfrac{A+B}{2}$.

Since $\cos\dfrac{A+B}{2}$ is real number, the above equation has a solution.

Thus, the discriminant $b^2 - 4ac \geq 0$, which gives

$\cos^2\dfrac{A-B}{2} - 8u \geq 0 \Rightarrow u \leq \dfrac{1}{8}\cos^2\dfrac{A-B}{2} \leq \dfrac{1}{8}$

Hence, $\sin\left(\dfrac{A}{2}\right)\sin\left(\dfrac{B}{2}\right)\sin\left(\dfrac{C}{2}\right) \leq \dfrac{1}{8}$

(iii) From (i) and (ii), we have $\cos A + \cos B + \cos C > 1$ and

$\cos A + \cos B + \cos C \leq 1 + 4 \times \dfrac{1}{8}$

Thus, we get $1 < \cos A + \cos B + \cos C \leq \dfrac{3}{2}$.

$\sin\dfrac{A}{2}\sin\dfrac{B}{2}\sin\dfrac{C}{2} > 0$, if $A + B + C = \pi$

Example 3.40 Prove that

$$\sin\frac{A}{2} + \sin\frac{B}{2} + \sin\frac{C}{2} = 1 + 4\sin\left(\frac{\pi - A}{4}\right)\sin\left(\frac{\pi - B}{4}\right)\sin\left(\frac{\pi - C}{4}\right), \quad \text{if } A + B + C = \pi$$

Solution:

$$\begin{aligned}
\text{L.H.S.} &= \cos\left(\frac{\pi}{2} - \frac{A}{2}\right) + \cos\left(\frac{\pi}{2} - \frac{B}{2}\right) + \cos\left(\frac{\pi}{2} - \frac{C}{2}\right) \\
&= \left[2\cos\left(\frac{\pi}{2} - \frac{A+B}{4}\right)\cos\left(\frac{B-A}{4}\right)\right] + \left[1 - 2\sin^2\left(\frac{\pi}{4} - \frac{C}{4}\right)\right] \\
&= 1 + 2\sin\left(\frac{\pi}{4} - \frac{C}{4}\right)\cos\left(\frac{B-A}{4}\right) - 2\sin^2\left(\frac{\pi}{4} - \frac{C}{4}\right) \\
&= 1 + 2\sin\left(\frac{\pi}{4} - \frac{C}{4}\right)\left[\cos\left(\frac{B-A}{4}\right) - \sin\left(\frac{\pi}{4} - \frac{C}{4}\right)\right] \\
&= 1 + 2\sin\left(\frac{\pi}{4} - \frac{C}{4}\right)\left[\cos\left(\frac{B-A}{4}\right) - \sin\left(\frac{A+B}{4}\right)\right] \\
&= 1 + 2\sin\left(\frac{\pi}{4} - \frac{C}{4}\right)\left[\cos\left(\frac{B-A}{4}\right) - \cos\left(\frac{\pi}{2} - \frac{A+B}{4}\right)\right] \\
&= 1 + 2\sin\left(\frac{\pi}{4} - \frac{C}{4}\right)\left[2\sin\left(\frac{\frac{B-A}{4} + \frac{\pi}{2} - \frac{A+B}{4}}{2}\right)\sin\left(\frac{\frac{\pi}{2} - \frac{A+B}{4} - \frac{B-A}{4}}{2}\right)\right] \\
&= 1 + 4\sin\left(\frac{\pi - A}{4}\right)\sin\left(\frac{\pi - B}{4}\right)\sin\left(\frac{\pi - C}{4}\right)
\end{aligned}$$